

BRINE EVOLUTION AND HABITABILITY IN ICE SHELLS ON OCEAN WORLDS: INSIGHTS FROM EXPERIMENTAL FREEZING AND GEOCHEMICAL MODELING. A. K. Schultz¹, A. Murdza¹, T. C. Tomlinson¹, and J. J. Buffo¹, ¹Thayer School of Engineering, Dartmouth College

Introduction: Several outer solar system moons have been designated as potentially habitable worlds. Icy satellites such as Europa and Enceladus are thought to have subsurface oceans, presenting compelling targets for solar system exploration [1]. In addition to the interior oceans of these icy moons, brine pockets within the ice shell of a body could be both favorable to life and more accessible than the ocean [2]. As such, both ocean and brine pocket chemistry are integral to the habitability prospects of an icy moon.

Recent work has shown that ion fractionation during ice formation in these ocean worlds can impact habitability conditions and the preservation of ocean chemistry within the ice shell [3]. By linking salt distributions to precipitation pathways and quantifying the degree of fractionation, we can interpret the chemistry of the ice shell and gain valuable information about the processes that shaped it, including whether ice shell chemistry on moons like Europa and Enceladus accurately reflects the chemistry of their subsurface oceans.

The continued freezing of brines trapped in the ice shell also plays a critical role in the habitability of the ice shell. As these brines freeze, they become isolated from the underlying ocean, potentially creating stable, nutrient-dense environments capable of supporting microbial life. Understanding how brines evolve during continued freezing within the ice shell provides insight into the possibility of long-term habitats in these frozen pockets.

Existing studies of icy moons and subsurface brine habitability rely on theoretical models of ice shell freezing. Here we benchmark the simulated freezing of brines against novel experimentally derived ice sample solutions. We explore how the evolution of brine pockets within an ice shell, frozen from an underlying ocean, influences ice mineralogy and its ability to support life. Additionally, we investigate the water properties of ionic strength, porosity, and water activity, which are critical for assessing habitability within these environments.

Methods: Eight ice cores of varying composition were grown at the Dartmouth Ice Research Laboratory via top-down solidification [4]. These cores were from compositions representing MgSO₄-rich, NaCl-rich, and Na₂CO₃-rich endmember solutions to account for putative estimates of the chemical compositions found on Europa and Enceladus. Each ice core was sliced in 2-centimeter vertical increments; the slices were subsequently melted to analyze the ionic composition

profile of the ice. The initial and final brine solution compositions were additionally analyzed.

We model the freezing pathways of these solutions using the geochemical modeling software PHREEQC [5]. We use the FREZCHEM database, a Pitzer model for freezing temperatures.

Freezing to in-situ temperatures We model the freezing of the sample solutions to the measured in-situ temperature of the ice core at depth. Throughout the experiments, Onset HOBO dataloggers tracked temperatures at the top of the ice core and in the underlying brine; we assume a linear temperature profile and calculate the temperature in the middle of each 2 cm sample. Upon simulated freezing, PHREEQC provides us with the residual brine composition, water properties (ionic strength, water activity, and porosity), and phase assemblage (including ice content) of the sample. We also model the freezing of the initial brine solution of each core to the same *in-situ* temperatures to compare the chemical composition and brine properties between the brines resulting from the freezing of the initial solution and the sample solutions at the same temperatures. Comparing the freezing evolution of the bulk initial solution to the already-frozen-and-melted sample solutions at the same temperatures is crucial for understanding how various thermal histories alter brine chemistry, water properties, and phase assemblages.

Freezing to eutectic temperatures We model the freezing pathways of the sample solutions from the in-situ ice core temperatures to progressively lower temperatures until we reach the eutectic, where no more liquid water remains. We track the phase assemblage, water properties, and composition of the remaining brine until the eutectic temperature; because brine volume fraction within ice is highly dependent on the salinity and composition of the ice, tracking these properties allows us to assess the stability of liquid water habitats in the ice shell [6].

Results and Discussion:

Freezing to in-situ temperatures We show the water properties of the residual brines derived from simulated freezing to in-situ temperatures in Figure 1. Water activity, which is directly correlated with temperature, increases at the same linear rate as temperature increases with depth. Ionic strength decreases with depth, particularly in cores with higher salinity endmembers, while porosity increases in the deeper regions. Together, these trends indicate that brine has not become as concentrated lower down in

the cores compared to the upper, colder regions. This implies that deeper brines may offer more stable and habitable conditions, with lower salinity and higher porosity potentially supporting microbial life by providing better access to nutrients and favorable ionic strength and water activity levels.

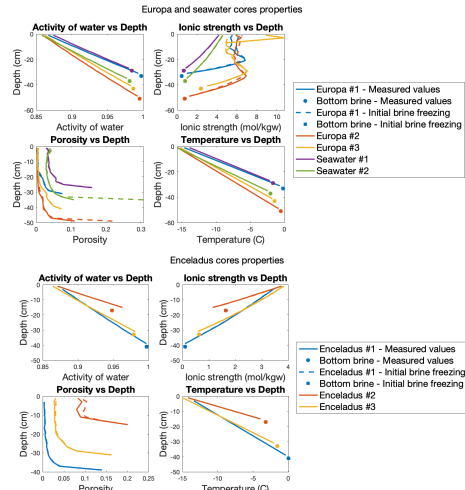


Figure 1. Residual brine properties (water activity, ionic strength, porosity, and temperature) for Europa, seawater, and Enceladus ice core samples. Solid lines indicate sample solution freezing and dashed lines indicate initial solution freezing.

Figure 2a shows the mineral assemblage at in-situ temperature for the Enceladus samples. During the freezing of the Enceladus core solutions, nahcolite precipitates due to the presence of bicarbonate in the endmember solution but decreases with depth, as the higher temperatures at the bottom of the core increase its solubility. This decrease in nahcolite precipitation with depth, along with lower ionic strength and higher porosity in the deeper regions, further indicates that the brine is not as concentrated in the lower parts of the cores compared to the upper, colder regions.

In contrast, the freezing of the Europa and seawater core solutions results in multiple precipitates, shown in Figure 2b. The precipitates of both resulting Europa and seawater brines include gypsum and mirabilite; meridianiite and picromerite only precipitate from the freezing of the Europa sample solutions. Gypsum suggests high calcium and sulfate concentrations, while meridianiite and mirabilite suggest high magnesium and sodium concentrations. Picromerite, however, only precipitates during the freezing of low-depth Europa #3 sample solutions. Picromerite formation requires high concentrations of potassium and magnesium, all of which are present within the top samples of Europa #3.

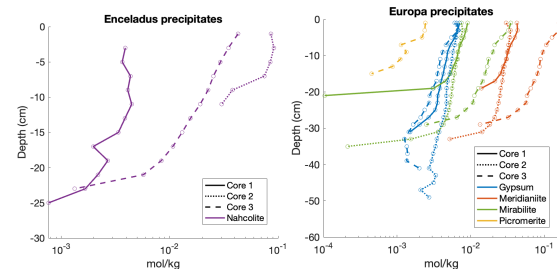


Figure 2. (a) phase assemblage of the residual brine at in-situ temperatures for Enceladus ice core samples. (b) phase assemblage of the residual brine at in-situ temperatures for Europa ice core samples.

Freezing to eutectic temperatures As a representative example of the work done, we show the evolution of the mineral assemblage due to continued freezing of the Europa #3 core sample solution at 17 cm depth in Figure 3. In addition to the already existing salts at in-situ temperature shown in Figure 2b, sylvite precipitates during continued freezing, with hydrohalite precipitation marking the real eutectic at approximately -32°C .

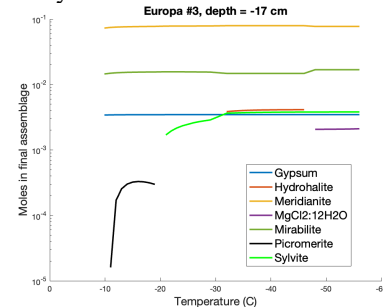


Figure 3. Evolution of mineral assemblage during freezing of Europa #3 at 17 cm depth. -32°C is the assumed real eutectic.

Conclusions: The water properties of the residual brines in the different cores provide crucial insights into how underlying ocean chemistries influence brine pocket habitability. For example, the high potassium concentration at the top of the Europa #3 core leads to an inhabitable spike in ionic strength, which could limit the potential for life, demonstrating key links between mineralogical assemblages, precipitation pathways, and habitability-relevant water properties. Additionally, our research shows that other factors like porosity are significantly affected by the ocean's chemistry, further shaping the conditions that could support or inhibit microbial life in these brine pockets.

References: [1] Soderlund, K. *et al.* (2020) Space Sci Rev, 216:80. [2] Marion, G. *et al.* (2003) Astrobio 3:4. [3] Wolfenbarger, N., *et al.* (2022) Astrobio 22:8. [4] Buffo, J. *et al.* (2023) LPSC 2023, abs. 1619 [5] Parkhurst, D., & Appelo, C. (2013) US Geological Survey. [6] Wolfenbarger, N. *et al.* (2022) GRL, 49.